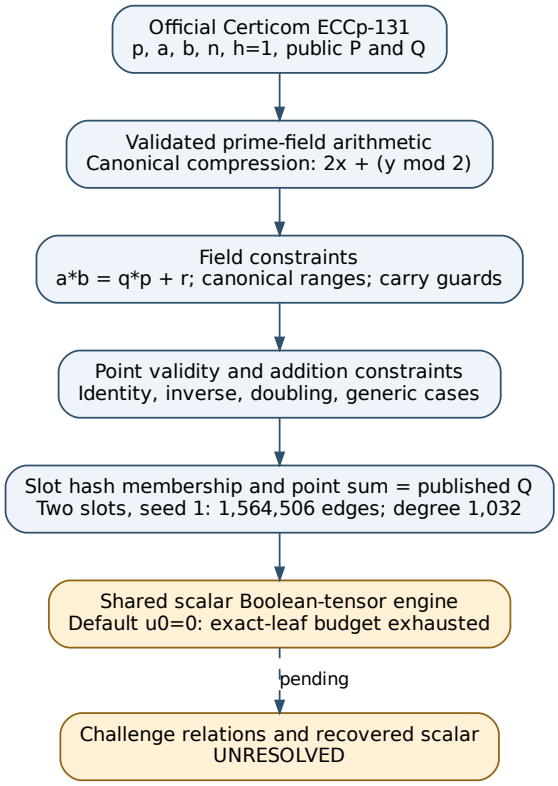
Certicom ECCp-131: scalar tensor reference

Implementation validation receipt

29 October 2026

Status: implemented reference; challenge scalar unresolved. The prime-field counterpart requested as "ECCp-130" is officially named ECCp-131 in Certicom's curve list. The implementation uses its exact published parameters and cofactor one. ECC2K-130 remains separately supported. Neither challenge is solved here.



The graph is a constraint pipeline, not an isogeny or a new curve route. Solid arrows are implemented composition; the dashed arrow is an unresolved challenge computation.

Construction and checks

The prime-field module implements short-Weierstrass arithmetic, Tonelli–Shanks decompression and canonical encoding $2x + (y \bmod 2)$. The coordinate module reuses the integer Fourier multiplier and enforces bounded integer equations $ab = qp + r$, canonical residues, addition/subtraction carries, exceptional point addition cases, fixed-cofactor image witnesses, affine hash thresholds and the target point sum. Test hints are absent from compiled tensor networks.

The combined suite passes 39 development tests with no skips: exhaustive modular pairs over $\mathbb{F}_{17}$, all 361 pairs in an order-19 point group, identity/inverse/zero-y/cofactor controls, invalid assignment rejection, certificate corruption rejection and independent OpenSSL comparisons of 16 multiples of the published prime-field P/Q. A separate public indexed-tensor fixture recovers scalar 7 and passes full relation-certificate replay. Producer and certificate verifier share arithmetic; OpenSSL supplies an independent arithmetic comparison, not a full proof of the tensor program.

The separately landed native Rust backend also passes its 27-test regression suite. The added published ECCp-131 runner input loads, but the bundled native budget stops exact AKS prime certification before tensor search, with no recovery receipt. This differs from the Python graph's later leaf limit.

The published target's two-slot, seed-1 network compiles to 1,564,506 edges, 654,523 variables and maximum degree 1,032. Compiled graph statistics agree with construction statistics. The default policy selects $u_0 = 0$; exact summation exceeds the 2,000,000-operation budget and exits 2 without a challenge witness. Enumerating its factor bases also fails the 200,000-candidate preflight. Neither failure proves nonexistence or a mathematical obstruction.

Scope and evidence

These are retrospective development checks, not a benchmark or admitted IC/rho comparison. Times in raw outputs are unisolated diagnostics. No new mathematical theorem, novelty, Lean proof, asymptotic speedup, native Rust backend port or scientific cloud experiment is claimed. A separate native tensor crate landed on main during this work; this change adds the published ECCp-131 runner input alongside it. The companion's general $\ell > 1$ case, independent complete-encoding, native-parity and complexity audits, and actual challenge relations/scalars remain open.

The companion directory retains commands, exit statuses, parameter/source hashes, environment, test output, fixture certificate and failed challenge receipts. The Markdown report (README.md) supplies exact reproduction commands, source links and acceptance limits. Diagram source is pipeline.dot. Existing native diagrams, reports and leaderboard rows are unchanged.

Sources: Certicom's official curve list; the pinned Bochum v0.2 paper; the external October 25 contraction snapshots listed in reference/SOURCES.json; the existing paired native reduction and canonical tensor-backend problem. No formal verification of the external theorem follows from these checks.